Licensed Electrician's Theory (LET) Licence Assessment
Sample Paper Marking guide (2021)

AS/NZS 3000 Wiring Rules

Q.1

Adjacent to but not on the water heater (2 marks)

Clause 4.8.2.3 (b) (2 marks)

Q.2

Normal operating conditions (2 marks)

Clause 1.7.1 (a) (2 marks)

Q.3

1. Installing the enclosure across the roofing profile or
2. Installing the enclosure within the valley or tray of the roofing material using supports that prevent the obstruction or water or accumulation of debris (2 marks) (either answer is acceptable)

Clause 3.10.3.1 (ii) or 3.10.3.1 (i) is acceptable. (2 marks)

Q.4

2.5 meters (2 marks)

Clause 6.6.3.4.2 (b) (2 marks)

AS/NZS 3012 Construction & Demolition Sites.

Q.5

40lx (2 marks)

Clause 2.7.1 (2 marks)

Q.6

A switch operating in all live (active and neutral) conductors (2 marks)

Clause 2.4.7 (c) (2 marks)

Electrical Safety (General) Regulations 2019

Q.7

The Apprentice Supervision Requirements (2 marks)

Clause 507 (1) (2 marks)

Electrical Shock Survival

Q.8

Place the casualty on his back (2 marks)
Tilt the head back and listen for breathing (2 marks)

Cable Selection

Q.9

Table 3(3) Item 4 (1 mark)

Table 14 Col 23 (2 marks, one for table and one for column)

Derating factor Table 25(2) 0.88 (1 mark)

Derating factor Table 28(1) 0.99 (1 mark)

Part (i) Answer: $16mm^2$ (2 marks)

Part (ii) Derating factor Table 25(2) 0.78

Answer: $25mm^2$ (1 mark)

(Deduct 1 mark for no or incorrect units on final answers. Only deduct one mark regardless of number of missing units)

DC Circuits

Q.10

(i) Meter X = ~~12.8W~~ (2 marks) 128W

(ii) Meter Y = 0.48A (2 marks)

(iii) Meter Z = 64V (2 marks)

(Deduct 1 Mark for no or incorrect units)

Maximum Demand

Q.11

Table C1 Column 2 (1 mark)

Domestic Residence.

Equipment	Load Group	Calculation	Maximum Demand
10A Double socket outlets **10A Single socket outlets** **34 Points total**	B(i)	10A for 1-20 points and 5A for each additional 20 points = 15A	15A (1 mark)
16 A electric vehicle charger	J(iv)	Full Connected Load 16A	16A (1 mark)
5 kw storage water heater	F	Full Connected Load 5000/230=21.74A	21.74A (1 mark)
Lights **36 Points Total**	A(i)	3A for 1-20 points and 2A for each additional 20 points = 5A	5A (1 mark)
3.3kW oven	C	50% connected load 3300/230= 14.35 x 0.5= 7.17A	7.17A (2 marks)
		Total Maximum Demand	64.91A (1 mark)

(Deduct 1 Mark for no or incorrect units on total, deduct 1 mark for no or incorrect load groups)

Voltage Drop

Q.12

Consumer Mains

Table 41 Column 10 (1 mark)

Vc 1.25 (1 mark)

Vd 2.56V (1 mark)

Sub-mains

Table 45 Column 6 (1 mark)

Vc 4.04 (1 mark)

Vd 11.27V (1 mark)

Final Sub-circuit

Table 41 Column 8 (1 mark)

Vc 4.05 (1 mark)

Vd 1.94V (1 mark)

Total Voltage Drop= 2.56 + 11.27 + 1.94 = 15.77V (1 mark)

(Deduct 1 Mark for no or incorrect units on total. Deduct 1 mark for no or incorrect table number/s)

Overload & Short Circuit Calculations

Q.13

Overcurrent divided by MCB current rating = 4 (1 mark)
Minimum Time = Accept 2-3 seconds (1 mark)
Maximum Time = Accept 6-8 seconds (1 mark)

(Deduct 1 mark for no or incorrect time unit)

Q.14

Transformer impedance

230/15,560 (2 marks)

0.01478Ω (1 mark) Answer must be to 5 decimal places.

Main switchboard prospective fault

230/ (0.01478 +0.0052) (2 marks)

11,511A (1 mark) (also accept 11,512A)

Distribution board prospective fault

230/ (0.01478 +0.0052+ 0.028) (2 marks)

4794A (1 mark)

(Deduct 1 Mark for no or incorrect units in final answer)

Residual Current Devices

Q.15

35A (1 Mark)

AS/NZS 3000 Clause 2.6.2.1 (a) (2 marks)

Motor and Starters

Q.16

D (2 Marks)

AS/NZS 4836:2011

Q.17

The general mass of earth at the work site (2 marks)
Clause number: ~~3.2.6~~ (2 marks) 3.1.6

Installation Defects- Non Domestic

Q.18

(2 marks for correct defect one mark for the correct clause)
(Only accept the first 5 defects candidate has listed)

1. - Main Neutral connection not indicated on neutral bar   2.10.5.4
2. - DB2 Isolator not rated for circuit voltage – 400V   2.2.4.2
3. - DB2 Fuses are undersized for MD load (Ib < In < Iz)   2.5.3.1
4. - Hot Water Service conductors undersize (T10, Col2 2.5mm2 = 27A) Min. size after 0.9 HRC derating = 27.78A   3.4.1 (no HRC fuse installed)
5. - DB2 Protective Earthing Conductor undersized for actives (should be 25mm2) 5.3.3.1.2 (a) Table 5.1
6. - Main Earth Electrode not driven to a minimum depth of 1.2M   5.3.6.3 (a)
7. - Final subcircuit neutrals do not correspond with final subcircuit actives (neutral sequencing)   2.10.5.4
8. - Hot Water Service Main Switch not marked "ON/OFF" positions 2.3.2.2.1 (c) ~~NOTE~~
9. - Main Switch does not indicate the portion of the installation it controls   2.3.3.5 (b)
10. - Location of Earth Electrode not indicated at Main Switchboard 5.3.6.4
11. ~~- Protective Earthing conductor for DB1 undersize should be 25mm2 for 70mm~~2 ~~submains - 5.3.3.1.2(a) Table 5.1~~
12. - DB1 fuses do not indicate what they control/protect   2.10.5.2
13. - No overload/short circuit protection at circuit origin for HWS 2.5.1.3
14. - Main Earthing conductor connection to Earth Electrode label (tag) does not state "Warning: Main Electrical Earthing Conductor – Do Not Disconnect" 5.5.1.3
15. - Main Earthing conductor is not coloured green/yellow 3.8.1 Table 3.4.~~1~~
16. - The Main Neutral, Main Earth and MEN connections are not under two screws or one screw with an outside diameter of not less than 80% of the tunnel diameter   2.10.4.2